ChatGPT

Institutional Catalog Underwriting Knowledge Base for Cadence

Executive summary

An institutional "underwriting engine" inside Cadence is not a separate product. It is **a deterministic math-and-controls layer** that sits on top of your existing accounting/royalty ingestion pipeline and produces (a) a **song-by-period revenue evolution table** (the spine), (b) **reconciled portfolio totals** (credibility), (c) **decay-based forward projections** (forecast), and (d) **valuation bands** (decision support).

To be underwriting-grade, the Knowledge Base (KB) must encode four things in a way the engine can ingest and replay:

1) **Canonical schemas** (works, recordings, parties, statements, line-items, FX, mappings, overrides, disputes/audits).
2) **Normalization rules** (identifiers, fuzzy matching, territory/currency, and periodization).
3) **Controls** (control totals, roll-forward balances, share integrity, three-way match patterns, and exception workflows).
4) **Analytics math** (decay fitting, half-life, volatility, concentration, scenario sensitivity) plus valuation templates (DCF and multiplier mappings).

Cadence should model statement transparency and traceability as if it were operating under the strictest available mechanical-royalty reporting requirements. Under US blanket mechanical streaming, regulations require royalty statements to include both the statement period and the activity period, and (critically) a **detailed step-by-step accounting of the royalty calculation** sufficient for validation; for adjustments the original period being adjusted must be clearly indicated. [1] This "explainability by construction" philosophy is what makes a valuation memo defensible.

The KB should also be standards-forward:
- ISWC is the ISO-standard identifier for musical works, intended to reduce ambiguity and improve rights administration efficiency. [2]
- ISRC is the ISO-standard identifier for sound recordings (ISO 3901), and is required/central in statutory reporting contexts like SoundExchange Reports of Use. [3]
- IPI is CISAC's system for globally unique party identification used in documentation/distribution/accounting across societies, administered by SUISA as IPI database administrator. [4]
- ISNI is ISO 27729 for public identities across media industries and is used as a disambiguation key in rights/admin contexts. [5]
- DDEX DSR and MWDR define practical, implementable vocabularies for usage/sales reporting and work rights share communications; DSR explicitly models territory and multi-currency reporting (ISO 4217) and exchange rates. [6]

Finally, **statement-only projections can be strong** if (i) your song-by-period table is clean, (ii) you separate earned vs cash (advances/recoupment/reserves), and (iii) you run sensitivity bands. Luminate later becomes a forecast “prior” (consumption trajectory), but statements remain the audit-proof ground truth for paid/owed amounts.

Canonical underwriting data model

Design objective

The KB’s data model must support three simultaneous “truths” without mixing them:

- **Earned** (activity-based): what the catalog generated in a given activity period.
- **Statement** (settlement-based): what the statement claims is payable for a statement period.
- **Cash** (posting-based): what was actually paid after recoupment/reserves/holds.

This separation mirrors regulated reporting expectations in MLC statements, which distinguish activity periods from statement periods and require explicit adjustment provenance. [1]

Canonical entities and key fields

Below is the minimal entity set to make underwriting reproducible and auditable.

Entity	Purpose	Primary keys	Must store
OrganizationTenant	Multi-tenancy boundary	`organization_id`	permissions, data isolation
Party	Any payee, rightsholder, society, DSP, label/publisher, admin	`party_id`	name, role, IPI/ISNI + provider IDs
Work	Composition	`work_id`	ISWC (nullable), titles, writer/publisher shares (time-valid)
Recording	Sound recording (master)	`recording_id`	ISRC (nullable), UPC, artists, versions
WorkRecordingLink	ISWC↔ISRC crosswalk	`link_id`	many-to-many edges + allocation rule
Statement	Document-level metadata	`statement_id`	issuer, currency, statement period, file refs, versioning
StatementLine	Atomic line item	`line_id`	work/recording links, activity period bounds, territory, source, income type, amounts

Entity	Purpose	Primary keys	Must store
AccountTransaction	Non-line flows affecting cash/ balances	`txn_id`	advance, recoupment, reserve, chargeback, payment, retro adjustment
FXRate	Conversion provenance	`(date, from, to, source)`	rate, rule (avg vs spot), applied fields
Mapping	ID normalization + fuzzy match results	`mapping_id`	confidence, evidence, override chain
Override	Manual correction with audit trail	`override_id`	reason, scope, effective dates, evidence refs
UnderwritingRun	“Projection button” snapshot	`run_id`	inputs, config version, outputs, exceptions

A key modeling insight from the USCO’s MLC database requirements: a robust rights system must treat “work” and “recording” as distinct objects and should carry both ISWC and ISRC plus data provenance for sound recording information. [7]

ER view

```
flowchart LR
  Party -->|owns/controls| Work
  Party -->|owns/controls| Recording
  Work --> WorkRecordingLink --> Recording
  Statement --> StatementLine
  StatementLine --> Work
  StatementLine --> Recording
  Statement --> AccountTransaction
  UnderwritingRun -->|reads| StatementLine
  UnderwritingRun -->|reads| AccountTransaction
  UnderwritingRun -->|writes| ValuationSnapshot
```

ETL, cleaning, identifier mapping, and overrides

ETL stages

```
flowchart TD
  A[Raw ingest\nPDF/CSV/XLSX/TSV/API] --> B[Immutable staging\nchecksum +
version]
  B --> C[Provider parser\nfield extraction]
  C --> D[Canonical normalize\nschema + types + signs]
  D --> E[ID mapping\nISWC/ISRC/IPI/ISNI + provider IDs]
  E --> F[Periodization\nactivity vs statement vs cash]
```

```
F --> G[Classification\nright + channel + flags]
G --> H[Controls\ncontrol totals + roll-forward]
H --> I[Underwriting views\nsong-by-period + portfolio]
I --> J[Decay + projection + valuation]
```

Identifier mapping and normalization methods

ISWC (works)
CISAC describes ISWC (ISO 15707) as a globally unique identifier for musical works that supports efficient worldwide rights administration and reduces ambiguity compared to title-only matching. [2]

ISRC (recordings)
IFPI notes ISRC is standardized as ISO 3901 and the ISRC system is designed so entities can obtain registrant codes and properly implement ISRC assignment. [8]

IPI (parties)
CISAC's IPI system exists for globally unique identification of interested parties and supports documentation, distribution, and accounting processes across member societies; SUISA administers the system. [4]

ISNI (public identities)
ISO specifies ISNI (ISO 27729) as a standard identifier to uniquely identify public identities across media content industries and disambiguate identities. [5]

Provider IDs
Provider statement systems often include internal work IDs, payee account codes, and deal scope IDs. The KB should treat these as "external IDs" with strong provenance and stable crosswalks.

Fuzzy-matching rules and scoring thresholds

Institutional underwriting does not tolerate "mystery matches." Every fuzzy match must have a score, and every score must be traceable.

A practical scoring rubric for Cadence:

Component	Method	Points
Title similarity	token-set ratio on normalized titles	0–60
Party overlap	Jaccard overlap of normalized writers/publishers/artists	0–30
Version evidence	duration, edit tags, release year match (if present)	0–10

Decision thresholds: - ≥ **90**: auto-link to canonical entity.
- **75–89**: "review required" queue.
- **< 75**: no link; remain unmatched/suspense.

Manual override workflows

Overrides must be their own records, not ad-hoc edits, because dispute policy ecosystems assume you can show provenance and change history. MLC policy framing emphasizes timely resolution among impacted parties and allows catalog-transfer conflict handling; that governance mindset is what you want your internal audit log to mirror. [9]

Override record fields:

Field	Why it matters
override_id, created_by, created_at	auditability
scope (work/recording/party/line)	avoids collateral damage
effective_start/end	supports time-valid mapping
reason_code + narrative	dispute-ready
evidence_refs	doesn't rely on memory
supersedes_mapping_id	chain of custody

Currency/territory normalization and FX policy templates

DDEX's DSR record definitions show how serious reporting systems treat currency and territory: territory is required and currency uses ISO 4217; exchange rate becomes mandatory when multiple currencies are reported. [10]

Cadence FX policy template (store in KB):

FX policy	Rule	Use case
Activity-average FX	average FX over activity range	earned-series analytics
Statement-date spot FX	spot on statement end date	settlement reporting
Cash-date spot FX	spot on cash posting date	cash forecasting
Reversal FX	reuse original FX rate when reversing	avoids artificial FX gains/losses

Territory normalization should map noisy labels to ISO codes with dictionary-first, then fuzzy:

Raw	Canonical
"US", "United States", "UNITED STATES"	US
"United Kingdom", "UNITED KINGDOM", "UK"	GB
"France", "FRANCE"	FR

Periodization rules and allocation of multi-month ranges

The KB must model three separate date constructs:

- `activity_start`, `activity_end` (earned window)
- `statement_start`, `statement_end` (statement coverage)
- `cash_posting_date` (when booked/paid)

This aligns with statutory reporting logic that explicitly requires reporting both statement period and activity period, and requires adjustment lines to carry original reporting period provenance. [1]

Allocation rules for multi-month ranges:

Scenario	Rule	Notes
Range aligns to full months	split evenly by month	default if no usage weights
Range overlaps partial months	prorate by days	common for mid-month licensing
Range is “settlement/true-up”	store as adjustment transaction + optional allocation	avoid contaminating earned decay unless explicitly allocated
Retro adjustment references original period	allocate to original period and tag as adjustment	mirrors MLC adjustment clarity expectations [11]

Classification taxonomy, reconciliation controls, audits, and disputes

Line-item classification taxonomy

Cadence should classify each line along four axes:

1) **Asset**: publishing-work vs master-recording
2) **Right type**: mechanical / performance / sync / print-lyrics / neighboring rights / other
3) **Channel**: streaming / download / broadcast / live / AV / UGC / social / physical / other
4) **Accounting flag**: current / retro adjustment / withholding / reserve movement / recoupment / chargeback

This is essential because regulated mechanical reporting (MLC) requires enough detail to verify rates and the payable royalty pool calculation; that implies your internal categories must be stable and replayable. [12]

Reconciliation algorithms

Institutional underwriting uses reconciliation as a credibility gate. If you can’t reconcile, your multiple compresses.

Control totals

For each statement:

- `Σ(gross_line_amount) = gross_payable`
- `Σ(withholding_tax) = withholding_total`
- `Σ(net_line_amount) = net_payable`

Balance roll-forward

When statements contain advances/recoupment balances, enforce:

`closing_balance = opening_balance + earned_royalties + adjustments − payments − new_advances ± other`

This is not optional: a catalog can “earn” money while paying nothing out due to recoupment.

Three-way match template

Check	Data sources	Pass condition
Statement totals vs line sums	statement header vs lines	exact (± rounding)
Line sums vs ledger entries	lines vs ledger postings	exact by period & currency
Ledger vs payment register/bank	ledger vs payments	exact by posting date

Audit and dispute workflows

Use the MLC ecosystem as a practical benchmark because it publicly documents audit mechanics.

- MLC DSP audits: initiated via Notice of Intent to Audit filed with the Copyright Office; MLC can audit a DSP no more than once per 3-calendar-year period; costs are covered via the MLC operational budget funded through DSP administrative assessment; audit recoveries are distributed without deducting audit costs. [13]
- MLC dispute policy framing: MLC dispute policies are intended to facilitate timely resolution among impacted parties, but MLC will not decide ownership questions; parties/courts decide. [9]

Cadence dispute workflow should mirror this logic:

```
flowchart TD
  A[User flags dispute\nwork/share/recording] --> B[Evidence attached\ncontracts, splits, statements]
  B --> C[Hold rule sets apply\nexclude from payable outputs]
  C --> D[Outreach + resolution\ninternal or counterparty]
  D --> E{Resolved?}
```

```
  E -->|Yes| F[Apply override\nre-run allocation + true-up]
  E -->|No| G[Escalate\nlegal/audit-ready packet]
```

Underwriting analytics: song-by-period spine, decay, concentration, and scenarios

The underwriting "spine" table

The most important table for institutional work is:

Song × Period × Revenue (by asset and right type)

Everything else (vintage analysis, DSP mix, portfolio projections) is derived from it.

Cadence output view (canonical):

song_id/work_id	period	publisher_net	writer_net	master_net	total_net

Decay analytics definitions

Cadence should compute decay per song and in aggregate, using both descriptive and model-based measures.

CAGR
For comparable start/end points (avoid post-hit spike vs long tail):

$$\mathrm{CAGR} = \left(\frac{y_T}{y_0}\right)^{\frac{1}{\Delta t}} - 1$$

Exponential decay (post-peak fit)

$$y_t = y_0 e^{-kt}$$

Half-life:

$$t_{1/2} = \ln(2)/k$$

Fit quality
Compute R^2 in log-space so you can punish bad fits:

$$R^2 = 1 - \frac{\sum(\ln y - \ln \hat{y})^2}{\sum(\ln y - \overline{\ln y})^2}$$

Volatility
Use the standard deviation of log returns for "stability":

$$\sigma = \mathrm{stdev}(\ln(r_t))$$

Concentration metrics

Must-compute concentration outputs: - Top-1 share, Top-3 share, Top-5 share, Top-10 share - HHI = $\sum s_i^2$

These are "multiple setters" because they measure dependence on one song.

Worked examples computed from uploaded statements

The following examples are computed from the uploaded statement CSV/PDF set associated with the publisher/admin statement periods 2022H2–2025H1, with work keys represented by `tango_work_code`. They are labeled as computed outputs, not external facts.

Statement-level royalty totals and withholding

Statement period	Gross paid	Withholding tax	Net paid less tax
2022-07-01..2022-12-31	17,092.67	-21.62	17,071.05
2023-01-01..2023-06-30	49,161.99	-33.43	49,128.56
2023-07-01..2023-12-31	47,970.07	-89.86	47,880.21
2024-01-01..2024-06-30	46,214.67	-76.91	46,137.77
2024-07-01..2024-12-31	41,057.85	-180.63	40,877.21
2025-01-01..2025-06-30	56,886.26	-62.66	56,823.60

Earned vs cash reality (advance/recoupment indicator)

Cover pages in the uploaded PDFs show negative closing balances across all statement periods (interpretable as unrecouped balance/advance position). This is exactly why Cadence must separate earned series from cash series.

Statement period	Closing balance (cover page)	Change vs prior
2022H2	-37,771.74	—
2023H1	-441,443.18	-403,671.44
2023H2	-395,336.22	+46,106.96
2024H1	-675,806.41	-280,470.19
2024H2	-639,684.20	+36,122.21

Statement period	Closing balance (cover page)	Change vs prior
2025H1	-539,369.17	+100,315.03

Interpretation: new advances or charges can make the balance more negative; earned royalties reduce the negative balance over time.

Song-by-period evolution table (statement periods)

Top works by total net (computed) shown across statement halves:

Work code	Work title	2022H2	2023H1	2023H2	2024H1	2024H2	2025H1
WW 015771185 00	PUSSY & MILLIONS	0.00	31,300.94	16,917.69	9,917.25	6,369.41	3,931.01
WW 010468147 00	HANDSOME & WEALTHY	4,825.13	5,834.72	6,799.41	10,352.75	6,909.36	9,842.17
WW 015189069 00	GUCCI FLIP FLOPS	2,083.48	1,731.68	2,189.49	3,746.88	3,946.47	7,731.76
WW 015108498 00	BARTIER CARDI	3,368.11	2,621.23	2,680.60	3,110.34	2,376.00	4,612.53
WW 015821643 00	RED RUBY DA SLEEZE	0.00	0.00	6,077.36	4,037.21	3,205.50	4,871.73

This is the exact “song-by-period” evolution construct used in institutional underwriting models.

Per-song decay fit example with overlay

Example work’s post-peak quarterly earned series (activity-quarter basis, net paid less tax), with peak-anchored exponential fit:

```
xychart-beta
  title "Decay overlay (Computed): PUSSY & MILLIONS (post-peak, quarterly)"
  x-axis
["2022Q4","2023Q1","2023Q2","2023Q3","2023Q4","2024Q1","2024Q2","2024Q3","2024Q4","2025Q1"]
  y-axis "Net (USD)" 0 --> 20000
  line
"Observed" [18179.79,10322.44,14958.47,6638.04,4025.31,8387.58,2372.96,1961.48,1219.43,370.15]
```

```
  line "Exp
fit"  [18179.79,12874.48,9117.38,6456.70,4572.48,3238.11,2293.15,1623.95,1150.04,814.43]
```

Computed decay parameters for this work: - $k \approx 0.345$ per quarter
- half-life ≈ 2.01 quarters (~6.0 months)

Portfolio half-life distribution (model output)

Bin counts of per-song half-life estimates (quarters) across decay-fit-eligible works:

```
xychart-beta
  title "Half-life histogram (Computed): per-song half-life in quarters"
  x-axis ["0-1q","1-2q","2-4q","4-8q","8q+"]
  y-axis "Works" 0 --> 25
  bar "Works" [23,20,16,8,3]
```

Concentration metrics by statement period

Statement period	Top-1 share	Top-3 share	Top-5 share	HHI
2022H2	0.283	0.614	0.796	0.160
2023H1	0.637	0.810	0.899	0.428
2023H2	0.353	0.622	0.767	0.179
2024H1	0.224	0.527	0.690	0.129
2024H2	0.180	0.504	0.679	0.111
2025H1	0.173	0.442	0.609	0.098

The 2023H1 concentration spike should be treated as a due-diligence flag: either it's a true superstar driver or a settlement/one-time event that must be normalized out.

Sensitivity scenarios (required by KB)

Cadence should always compute scenario outputs at least across these toggles:

Scenario toggle	Why it matters
Earned vs statement vs cash	prevents recoupment distortion
Include vs exclude sync and print/lyrics	lumpy revenue can inflate multiples
Gross vs net-of-withholding	controls tax distortion
Peak-anchored vs rolling-window decay	avoids "one spike defines the curve"

Master vs publishing reconciliation and statement parsers

Master vs publishing reconciliation (ISRC↔ISWC crosswalk)

The USCO's MLC database rule explicitly requires identifying sound recordings in which a work is embodied, and includes ISRC among the relevant fields, which is a strong external validation that a crosswalk is not optional for modern reconciliation. [14]

Cadence reconciliation policies should default to:

Case	Allocation rule	When to require manual
1 work ↔ many recordings	allocate publishing to work; allocate masters to recording; compare at aggregate	only if missing links
1 recording ↔ many works	allocate master revenue by recording; allocate publishing by linked work weights	if medleys/samples unclear
Missing ISRC/ISWC	use title/artist/party fuzzy with review thresholds	always for high-value lines

Sample statement parsers and expected fields

MLC statement parser (TSV)

MLC provides statements in TSV and allows export in Work Summary or Royalty Detail layouts; Royalty Detail corresponds to individual sound recordings within DSP usage reports, while Work Summary is aggregated at work level. [15]

Regulatory requirements include reporting both statement and activity period and providing a step-by-step accounting of royalty calculation under the CRB rate rules (Part 385). [12]

Minimum canonical mapping:

MLC field family	Canonical target
payee identifiers, IPI/ISNI	Party.external_ids
work identifiers (ISWC)	Work.iswc
recording identifiers (ISRC)	Recording.isrc
payable units, use type, service tier	StatementLine.units/channel
calc components (pool, rate, interest)	StatementLine.calculation_json

PRO parser pattern (BMI emphasized)

BMI registrations are on a 200% scale: 100% writer share and 100% publisher share; if no publisher is indicated, BMI may revert publisher share to the writer. BMI also states that CAE/IPI numbers can be found via Songview and are shown on statements. [16]

Minimum canonical mapping:

PRO field family	Canonical target
work title + PRO work ID	Work.title + Work.external_id(PRO)
payee CAE/IPI	Party.external_ids (IPI)
distribution quarter + category	StatementLine.activity_period (quarter)
writer vs publisher splits	RightsClaim/share integrity checks

SoundExchange parser (statutory RU)

SoundExchange reporting requirements emphasize that statutory licensees must submit Reports of Use with detailed track info plus audience measurement, and identify the recording using ISRC (or specified alternatives). [17]

Minimum canonical mapping:

RU field family	Canonical target
service provider + reporting category	Statement + StatementLine.source_provider
track title, artist, ISRC	Recording.isrc + Recording.title + artist
total performances / ATH	units + measurement_type
reporting month	activity period (month)

Valuation module: DCF, multipliers, and tax notes

DCF template

Cadence should store forecast assumptions as first-class inputs. If your mechanical streaming forecast is used, your KB should reference that the underlying statutory streaming mechanical pool calculation is formula-driven and relies on percent-of-revenue and TCC constructs. [18]

DCF core:

- Build a forecast per song (or song bucket) for 5–10 years.
- Apply a terminal value (exit multiple or perpetual growth).
- Discount using scenario discount rates.

Year	Forecast net	Decay/growth	Discount rate	PV
1	N_1	g_1	r	PV_1
…	…	…	…	…
Terminal	N_T × multiple	—	r	PV_T

Income-multiplier mapping

Multiples should not be a single number. The KB should encode a mapping from stability signals to suggested multiple bands, for example:

- Higher multiple: low volatility, diversified territories, diversified DSPs, low concentration.
- Lower multiple: high concentration, large one-time sync, unresolved disputes, missing statement periods, heavy recoupment drag.

Tax/amortization note (buyer-side)

In the US, the IRS explains that acquired "section 197 intangibles" are generally amortized over 15 years when held for business/income production, which affects buyer after-tax valuation models. [19]

Delivery-ready KB machine pack

The following JSON and YAML packages are deliberately structured so Cadence can load them as a versioned KB (e.g., `kb/underwriting_kb.json` or `kb/underwriting_kb.yaml`) and the underwriting engine can run the same logic for every "projection button" invocation.

JSON package


```json
{
  "kb_name": "cadence_institutional_underwriting_kb",
  "kb_version": "2026-02-27",
  "principles": {
    "separate_time_axes": ["activity", "statement", "cash"],
    "earned_vs_cash": "Earned series is activity-based; cash series applies advances/reserves/recoupment and payment postings.",
    "immutability": "Raw files are immutable; all canonical transforms are reproducible and versioned.",
    "provenance_required": true
  },
  "enums": {
    "asset_type": ["publishing_work", "master_recording"],
    "right_type": ["mechanical", "performance", "sync", "print_lyrics", "neighboring_rights", "other"],
    "channel": ["streaming", "download", "broadcast", "live", "av", "ugc", "social", "physical", "other"],
```

```
    "accounting_flag": ["current", "retro_adjustment", "withholding",
"reserve_movement", "recoupment", "chargeback", "hold"],
    "currency_code": "ISO4217",
    "territory_code": "ISO3166_1_alpha2"
  },
  "schemas": {
    "Statement": {
      "required": ["statement_id", "issuer_name", "statement_period",
"statement_currency", "source_type"],
      "fields": {
        "statement_id": "string",
        "organization_id": "string",
        "issuer_name": "string",
        "source_type": "publisher_admin|pro|mlc|soundexchange|label_distributor|
other",
        "statement_period": {"start": "date", "end": "date"},
        "statement_currency": "ISO4217",
        "statement_version": "string|null",
        "raw_files": ["string"],
        "totals": {
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          "withholding_tax": "number|null",
          "net_payable": "number|null"
        },
        "balances": {
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          "closing_balance": "number|null",
          "amount_due": "number|null"
        }
      }
    },
    "StatementLine": {
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"activity_period"],
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        "statement_id": "string",
        "asset_type": "asset_type",
        "work_id": "string|null",
        "recording_id": "string|null",
        "provider_work_id": "string|null",
        "provider_recording_id": "string|null",
        "activity_period": {"start": "date|null", "end": "date|null",
"year_month": "string|null"},
        "territory_raw": "string|null",
        "territory": "territory_code|null",
        "source": {"payer": "string|null", "dsp": "string|null", "society":
"string|null"},
```

```
        "right_type": "right_type",
        "channel": "channel",
        "accounting_flags": ["accounting_flag"],
        "units": "number|null",
        "rate_pct": "number|null",
        "amount_gross": "number",
        "withholding_tax": "number|null",
        "amount_net": "number|null",
        "currency": "ISO4217",
        "fx": {"policy": "string|null", "rate": "number|null", "from": "ISO4217|null", "to": "ISO4217|null"},
        "provenance": {"source_file": "string", "source_row": "number|null", "source_page": "number|null"}
      }
    },
    "Mapping": {
      "required": ["mapping_id", "mapping_type", "from_key", "to_key", "confidence"],
      "fields": {
        "mapping_id": "string",
        "organization_id": "string",
        "mapping_type": "work|recording|party",
        "from_key": "string",
        "to_key": "string",
        "match_method": "exact_id|exact_normalized|fuzzy|manual_override",
        "confidence": "number",
        "evidence_refs": ["string"],
        "effective_start": "date|null",
        "effective_end": "date|null",
        "supersedes_mapping_id": "string|null"
      }
    },
    "UnderwritingRun": {
      "required": ["run_id", "organization_id", "scope", "kb_version", "created_at"],
      "fields": {
        "run_id": "string",
        "organization_id": "string",
        "scope": {"creator_id": "string|null", "catalog_id": "string|null"},
        "kb_version": "string",
        "created_at": "datetime",
        "inputs": {
          "periodization_mode": "activity|statement|cash",
          "exclude_right_types": ["right_type"],
          "exclude_flags": ["accounting_flag"],
          "fx_policy": "string",
          "maturity_cutoff_quarters": "number"
        },
```

```
        "outputs": {
          "song_by_period_table_ref": "string",
          "portfolio_summary_ref": "string",
          "decay_params_ref": "string",
          "projection_ref": "string",
          "valuation_ref": "string",
          "exceptions_ref": "string"
        }
      }
    }
  },
  "normalization": {
    "territory_dictionary": {
      "US": ["US", "United States", "UNITED STATES", "USA"],
      "GB": ["GB", "United Kingdom", "UNITED KINGDOM", "UK"],
      "FR": ["FR", "France", "FRANCE"],
      "CA": ["CA", "Canada", "CANADA"]
    },
    "title_normalization": "upper_trim_strip_punct_collapse_ws",
    "party_normalization": "upper_trim_strip_punct",
    "currency_policy": {
      "base_currency": "configurable_per_organization",
      "fx_timestamp_rule_default": "average_over_activity_period",
      "reversal_fx_rule": "reuse_original_fx"
    }
  },
  "fuzzy_matching": {
    "weights": {"title": 60, "parties": 30, "version": 10},
    "thresholds": {"auto": 90, "review": 75},
    "gap_safe_ratio": "per_period_ratio = (y_t / y_prev)^(1/gap_periods)"
  },
  "classification": {
    "right_type_rules": [
      {"if_income_type_contains": ["Synch", "Synchronisation"], "right_type":
"sync"},
      {"if_income_type_contains": ["Print", "Lyrics"], "right_type":
"print_lyrics"},
      {"if_income_type_contains": ["Mechanical"], "right_type": "mechanical"},
      {"if_income_type_contains": ["Performance"], "right_type": "performance"}
    ],
    "channel_rules": [
      {"if_source_contains": ["Spotify", "Apple", "YouTube", "TikTok"],
"channel": "streaming"},
      {"if_income_type_contains": ["Download"], "channel": "download"},
      {"if_income_type_contains": ["Radio", "TV", "Film"], "channel":
"broadcast"},
      {"if_income_type_contains": ["Live"], "channel": "live"}
    ],
```

```
    "flag_rules": [
      {"if_text_contains": ["ADJUST", "CORRECTION", "RETRO"], "flag":
"retro_adjustment"},
      {"if_text_contains": ["WITHHOLD", "TAX"], "flag": "withholding"},
      {"if_text_contains": ["RESERVE"], "flag": "reserve_movement"},
      {"if_text_contains": ["RECOUP"], "flag": "recoupment"}
    ]
  },
  "controls": {
    "control_totals": [
      "sum(amount_gross) == Statement.totals.gross_payable",
      "sum(withholding_tax) == Statement.totals.withholding_tax",
      "sum(amount_net) == Statement.totals.net_payable"
    ],
    "share_integrity": [
      "sum(work_share_pct by work/right/territory) == 100%"
    ],
    "three_way_match": [
      "statement_totals == line_sums",
      "line_sums == ledger_entries",
      "ledger_entries == payment_register"
    ]
  },
  "analytics": {
    "decay": {
      "exp_fit": "y_t = y0 * exp(-k*t)",
      "k_estimator": "k = -sum(t * ln(y/y0)) / sum(t^2)",
      "half_life": "ln(2)/k",
      "r2_log": "1 - SSE(log(y),log(yhat))/SST(log(y))",
      "volatility": "stdev(log(per_period_ratio))"
    },
    "concentration": {
      "top_n": [1, 3, 5, 10],
      "hhi": "sum(share^2)"
    },
    "scenario_matrix": {
      "exclude_sync_print": true,
      "maturity_cutoff_quarters_default": 2,
      "decay_bands": {"downside": 1.25, "base": 1.0, "upside": 0.75}
    }
  },
  "valuation": {
    "methods": ["multiplier", "dcf"],
    "dcf": {
      "horizon_years": 10,
      "terminal_method": "exit_multiple|gordon_growth",
      "discount_rate_bands": {"low": 0.09, "base": 0.11, "high": 0.14}
    },
```

```
    "multipliers": {
      "publishing_nps": {"low": 10, "base": 13, "high": 16},
      "masters_net": {"low": 6, "base": 9, "high": 12}
    },
    "tax_notes": {
      "us_section_197": "15-year amortization for acquired section 197
intangibles (buyer-side modeling)."
    }
  }
}
```


YAML package


```
kb_name: cadence_institutional_underwriting_kb
kb_version: "2026-02-27"

principles:
  separate_time_axes: [activity, statement, cash]
  earned_vs_cash: "Earned series is activity-based; cash series applies
advances/reserves/recoupment and payment postings."
  immutability: "Raw files are immutable; canonical transforms are reproducible
and versioned."
  provenance_required: true

enums:
  asset_type: [publishing_work, master_recording]
  right_type: [mechanical, performance, sync, print_lyrics, neighboring_rights,
other]
  channel: [streaming, download, broadcast, live, av, ugc, social, physical,
other]
  accounting_flag: [current, retro_adjustment, withholding, reserve_movement,
recoupment, chargeback, hold]
  currency_code: ISO4217
  territory_code: ISO3166_1_alpha2

schemas:
  Statement:
    required: [statement_id, issuer_name, statement_period, statement_currency,
source_type]
    fields:
      statement_id: string
      organization_id: string
      issuer_name: string
      source_type: "publisher_admin|pro|mlc|soundexchange|label_distributor|
other"
      statement_period: {start: date, end: date}
```

```
      statement_currency: ISO4217
      statement_version: string|null
      raw_files: [string]
      totals:
        gross_payable: number|null
        withholding_tax: number|null
        net_payable: number|null
      balances:
        opening_balance: number|null
        closing_balance: number|null
        amount_due: number|null

  StatementLine:
    required: [line_id, statement_id, amount_gross, currency, activity_period]
    fields:
      line_id: string
      statement_id: string
      asset_type: asset_type
      work_id: string|null
      recording_id: string|null
      provider_work_id: string|null
      provider_recording_id: string|null
      activity_period: {start: date|null, end: date|null, year_month: string|
null}
      territory_raw: string|null
      territory: territory_code|null
      source: {payer: string|null, dsp: string|null, society: string|null}
      right_type: right_type
      channel: channel
      accounting_flags: [accounting_flag]
      units: number|null
      rate_pct: number|null
      amount_gross: number
      withholding_tax: number|null
      amount_net: number|null
      currency: ISO4217
      fx: {policy: string|null, rate: number|null, from: ISO4217|null, to:
ISO4217|null}
      provenance: {source_file: string, source_row: number|null, source_page:
number|null}

  Mapping:
    required: [mapping_id, mapping_type, from_key, to_key, confidence]
    fields:
      mapping_id: string
      organization_id: string
      mapping_type: "work|recording|party"
      from_key: string
```

```yaml
      to_key: string
      match_method: "exact_id|exact_normalized|fuzzy|manual_override"
      confidence: number
      evidence_refs: [string]
      effective_start: date|null
      effective_end: date|null
      supersedes_mapping_id: string|null

  UnderwritingRun:
    required: [run_id, organization_id, scope, kb_version, created_at]
    fields:
      run_id: string
      organization_id: string
      scope: {creator_id: string|null, catalog_id: string|null}
      kb_version: string
      created_at: datetime
      inputs:
        periodization_mode: "activity|statement|cash"
        exclude_right_types: [right_type]
        exclude_flags: [accounting_flag]
        fx_policy: string
        maturity_cutoff_quarters: number
      outputs:
        song_by_period_table_ref: string
        portfolio_summary_ref: string
        decay_params_ref: string
        projection_ref: string
        valuation_ref: string
        exceptions_ref: string

normalization:
  territory_dictionary:
    US: [US, "United States", "UNITED STATES", USA]
    GB: [GB, "United Kingdom", "UNITED KINGDOM", UK]
    FR: [FR, France, FRANCE]
    CA: [CA, Canada, CANADA]
  title_normalization: upper_trim_strip_punct_collapse_ws
  party_normalization: upper_trim_strip_punct
  currency_policy:
    base_currency: configurable_per_organization
    fx_timestamp_rule_default: average_over_activity_period
    reversal_fx_rule: reuse_original_fx

fuzzy_matching:
  weights: {title: 60, parties: 30, version: 10}
  thresholds: {auto: 90, review: 75}
  gap_safe_ratio: "per_period_ratio = (y_t / y_prev)^(1/gap_periods)"
```

```yaml
classification:
  right_type_rules:
    - if_income_type_contains: ["Synch", "Synchronisation"]
      right_type: sync
    - if_income_type_contains: ["Print", "Lyrics"]
      right_type: print_lyrics
    - if_income_type_contains: ["Mechanical"]
      right_type: mechanical
    - if_income_type_contains: ["Performance"]
      right_type: performance
  channel_rules:
    - if_source_contains: ["Spotify", "Apple", "YouTube", "TikTok"]
      channel: streaming
    - if_income_type_contains: ["Download"]
      channel: download
    - if_income_type_contains: ["Radio", "TV", "Film"]
      channel: broadcast
    - if_income_type_contains: ["Live"]
      channel: live
  flag_rules:
    - if_text_contains: ["ADJUST", "CORRECTION", "RETRO"]
      flag: retro_adjustment
    - if_text_contains: ["WITHHOLD", "TAX"]
      flag: withholding
    - if_text_contains: ["RESERVE"]
      flag: reserve_movement
    - if_text_contains: ["RECOUP"]
      flag: recoupment

controls:
  control_totals:
    - "sum(amount_gross) == Statement.totals.gross_payable"
    - "sum(withholding_tax) == Statement.totals.withholding_tax"
    - "sum(amount_net) == Statement.totals.net_payable"
  share_integrity:
    - "sum(work_share_pct by work/right/territory) == 100%"
  three_way_match:
    - "statement_totals == line_sums"
    - "line_sums == ledger_entries"
    - "ledger_entries == payment_register"

analytics:
  decay:
    exp_fit: "y_t = y0 * exp(-k*t)"
    k_estimator: "k = -sum(t * ln(y/y0)) / sum(t^2)"
    half_life: "ln(2)/k"
    r2_log: "1 - SSE(log(y),log(yhat))/SST(log(y))"
    volatility: "stdev(log(per_period_ratio))"
```

```yaml
  concentration:
    top_n: [1, 3, 5, 10]
    hhi: "sum(share^2)"
  scenario_matrix:
    exclude_sync_print: true
    maturity_cutoff_quarters_default: 2
    decay_bands: {downside: 1.25, base: 1.0, upside: 0.75}

valuation:
  methods: [multiplier, dcf]
  dcf:
    horizon_years: 10
    terminal_method: "exit_multiple|gordon_growth"
    discount_rate_bands: {low: 0.09, base: 0.11, high: 0.14}
  multipliers:
    publishing_nps: {low: 10, base: 13, high: 16}
    masters_net: {low: 6, base: 9, high: 12}
  tax_notes:
    us_section_197: "15-year amortization for acquired section 197 intangibles (buyer-side modeling)."
```

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[4] IPI | CISAC
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https://www.soundexchange.com/service-provider/reporting-requirements/

[18] https://www.law.cornell.edu/cfr/text/37/385.21
https://www.law.cornell.edu/cfr/text/37/385.21

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https://www.irs.gov/businesses/small-businesses-self-employed/intangibles